

INFORMATIONAL ADVANTAGES OF COORDINATE SYSTEMS: PHASE TRANSITIONS AND TOPOLOGY

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ABSTRACT. We compare entropy functionals on objects in a category to quantify how effectively different decompositions—such as bases or coordinate systems—concentrate information. This comparison partitions the category into phases where one decomposition systematically dominates another. We provide a simple criterion ensuring these phases form sieves and cosieves, stable under pre- or post-composition of morphisms. We illustrate the framework concretely on quantum states and polynomial decompositions, providing a general categorical method for comparing coordinate systems through information theory.

1. INTRODUCTION

We introduce a categorical framework to compare entropy functionals, showing how these comparisons yield well-defined phase transitions that measure how effectively different decompositions—such as coordinate systems—concentrate or disperse information.

Let $\mathcal{C}$ be a category. Given two scalar assignments on objects $F, G : \text{Ob}(\mathcal{C}) \rightarrow \mathbb{R}$, define the **indicator**

$$\delta_G^F(X) := F(X) - G(X), \quad \delta_G^F(f : X \rightarrow Y) := \delta_G^F(Y) - \delta_G^F(X).$$

and study the partition into **phases** A and B that δ_G^F induces on a category defined as

$$A := \{X \mid \delta_G^F(X) \leq 0\}, \quad B := \{X \mid \delta_G^F(X) \geq 0\}.$$

When $\delta_G^F(f) \geq 0$ for all f , the phases A and B become **closed** under precomposition and composition respectively. That is, A generates a sieve and B a cosieve.

We prove this result in Section 2 and present two case studies with **closed phase transitions** in Section 3; one in quantum mechanics and one in polynomial algebra.

1.1. Motivation. Families of entropies (von Neumann, Rényi, Tsallis) are routinely used to probe different informational effects in quantum information and mathematical physics [CV19]. Entropic uncertainty principles [FGG13] and inter-entropy bounds [SNP23] appear in our setting as *phases*. While there is a rich literature axiomatizing and functorially characterizing specific entropies [Fad57, BFL11, Bra21, Del19], our focus here is orthogonal: we study the *topology* induced by comparing two functionals across a category.

The many powerful properties of entropy allow for closed phase transitions to manifest and be proven. Results such as the data-processing inequality [CV19, FGG13], Schur convexity, and majorization yield tractable bounds along morphisms and practical criteria ensuring the closed-phase conditions.

Operationally, entropy quantifies information concentration for a chosen decomposition (spectrum, basis, coordinates). Comparing two entropies thus compares *coordinate systems*: low values indicate focused information, high values dispersed information. The indicator δ_G^F then measures the objectwise *information advantage* of one decomposition over another and—under our hypotheses—a change between which coordinate system is more effective can occur at most once along a chain of morphisms.

2. PHASE TOPOLOGY

Throughout, F and G denote scalar assignments on objects of a category $\mathcal{C}$. Recall the indicator δ_G^F and the corresponding phases A and B . The intersecting region $A \cap B$, where the two assignments are equal, is called the **phase boundary**. If neither A nor B is equal to the boundary $A \cap B$, we say that $\mathcal{C}$ exhibits a **phase transition**, since there exist objects $X, Y \in \text{Ob}(\mathcal{C})$ with $\delta_G^F(X) < 0$ and $\delta_G^F(Y) > 0$.

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Proposition 2.1. *Let $\mathcal{C}$ be a category, and let (F, G) be scalar functionals. If for all arrows $f : X \rightarrow Y$, $\delta_G^F(f) \geq 0$, then A and B generate a sieve and cosieve respectively.*

Proof. For the sieve claim, suppose $Y \in A$, i.e., $\delta_G^F(Y) \leq 0$, and $\delta_G^F(f) \geq 0$. Then:

$$\delta_G^F(X) = \delta_G^F(Y) - \delta_G^F(f) \leq 0 - 0 = 0,$$

so $X \in A$.

For the cosieve claim, suppose $X \in B$, i.e., $\delta_G^F(X) \geq 0$, and $\delta_G^F(f) \geq 0$. Then:

$$\delta_G^F(Y) = \delta_G^F(X) + \delta_G^F(f) \geq 0 + 0 = 0,$$

so $Y \in B$. □

The sieve structure ensures persistence of phase membership: along any composable chain, once an object enters a phase it cannot flip back to the other (a transition can occur at most once along the chain). We call such a transition **closed**.

3. CASE STUDIES

We present two case studies. First, we reinterpret the Heisenberg uncertainty principle within our framework, showing how position and momentum decompositions yield complementary informational advantages and exhibit a phase transition. Then, we construct an analogue of a noisy channel in polynomial algebra, where expanded and factored forms of polynomials dominate in different regions, also exhibiting a phase transition.

3.1. Solitonic Quantum States. Consider the symmetric potential well with width λ and constant V_0 :

$$V_\lambda(x) = V_0 \frac{1 - \lambda x \cot(\lambda x)}{(\lambda x)^2}.$$

Next, define two entropy assignments on quantum states, the position entropy S_x and the momentum entropy S_p :

$$S_x(\psi) = - \int |\psi(x)|^2 \ln |\psi(x)|^2 dx,$$

$$S_k(\psi) = - \int |\phi(k)|^2 \ln |\phi(k)|^2 dk, \quad \phi(k) = \frac{1}{\sqrt{2\pi}} \int e^{-ikx} \psi(x) dx.$$

Let $\psi_n(\lambda)$ denote the solitonic (time independent, localized, finite energy) quantum states which solve the following Schrödinger equation:

$$H\psi = \frac{-1}{2} \frac{d\psi}{dx^2} + V(\lambda, x)\psi.$$

Recent work by [ORU25, Lim22] has highlighted an interesting manifestation of the Heisenberg uncertainty principle, that for these such states,

$$(3.1) \quad \frac{d}{d\lambda}(S_x(\psi_n(\lambda)) + S_k(\psi_n(\lambda))) = 0.$$

Define a category **State_V** to have objects (λ, n) , where n denotes the quantum number and λ the width parameter. Each object is a state $\psi_n(\lambda)$. Morphisms are the thin poset defined through the partial order

$$(\lambda, n) \rightarrow (\lambda', n') \iff n = n' \text{ and } \lambda \leq \lambda'.$$

Corollary 3.1. *The pair (S_x, S_k) exhibits a **closed phase transition** on **State_V**. The phase transition for the ground state $\psi_0(\lambda)$ is visualized in Figure 1.*

Proof. We will first show the condition $\delta_k^x(f) \geq 0$ and then construct a strict overlap to show that $A \neq A \cap B$ and $B \neq A \cap B$.

Equation 3.1 directly implies that $\delta_k^x(f) \geq 0$. Since morphisms are increases in the parameter λ , S_x and S_k can only move in opposite directions morphisms, widening the gap δ_G^F .

To see the strict overlap, choose objects such that $\lambda_1 < 1.66$ and $\lambda_2 > 1.66$. Numerical computations in Figure 1 confirm overlapping behavior for $n = 0$, that is

$$\delta_k^x(\psi_0(\lambda_1)) < 0, \quad \delta_k^x(\psi_0(\lambda_2)) > 0.$$

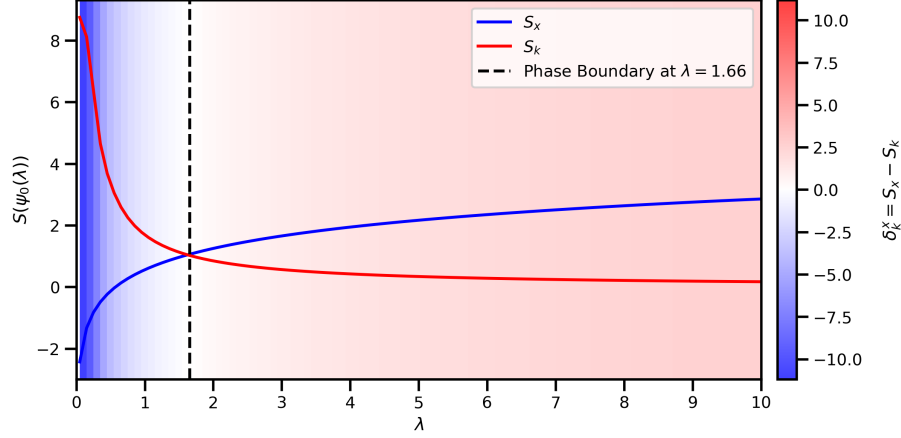


FIGURE 1. Phase transition in the ground states $\psi_0(\lambda)$ for solitonic solutions of the Schrödinger equation with potential $V(\lambda)$. Blue indicates position dominance, and red indicates momentum dominance, transitioning clearly at $\lambda = 1.66$. Data simulated using equations from [ORU25].

□

3.2. Polynomials. Fix a complete ground field $K \subseteq \mathbb{C}$ (take $K = \mathbb{C}$ for simplicity) and set

$$\mathbf{Poly}_\times := \{\text{monic, split polynomials } p \in K[y]\},$$

$$\text{Hom}_{\mathbf{Poly}_\times}(p, q) := \left\{ M_r : p \mapsto pr \mid \begin{array}{l} r \text{ monic, square-free,} \\ \text{roots of } r \text{ disjoint from those of } p \end{array} \right\}.$$

The disjointness condition ensures that morphism composition is well-defined: if r and r' are square-free with disjoint support from the roots of p and pr respectively, then their product rr' is also square-free and has disjoint support from the roots of p . Thus, $M_{r'} \circ M_r = M_{rr'}$ is a valid morphism, and $\mathbf{Poly}_\times$ forms a category. Morphisms here model a noisy channel for polynomials which guarantee increased complexity.

Notice the two equivalent representations of a polynomial,

$$p(y) = \prod_{j=1}^n (y - r_j)^{m_j} = \sum_{k=0}^n c_k y^k.$$

Using these, define two entropy functionals

$$S_r(p) = - \sum_{j=1}^n \hat{m}_j \log(\hat{m}_j), \quad S_c(p) = - \sum_{k=0}^n \hat{c}_k \log(\hat{c}_k),$$

where $\hat{m}_j$ and $\hat{c}_k$ are the normalized (and positive) multiplicities and coefficients respectively.

In order to achieve the monotonicity condition $\delta(f) \leq 0$, that is $[S_r(pr) - S_r(p)] \geq [S_c(pr) - S_c(p)]$, we require the additional assumption that the roots of r are not extremely far from the roots of p .

Proposition 3.2. *Let $f : p \rightarrow rp$ be a morphism in $\mathbf{Poly}_\times$, and write*

$$p = \prod_{i=1}^n (x - a_i)^{f_i} \quad \text{and} \quad r = \prod_{j=1}^k (x - b_j).$$

Let $m = \max |a_i|$ and $C > 0$. If $b_j < Cm$ for all j , then for every degree pair (n, k) with $n + k \geq N(C)$ where $N(C)$ is the unique integer solving

$$\frac{\ln N}{N} = \ln(1 + C),$$

the monotonicity condition $\delta_c^r(f) \geq 0$ holds.

Proof. We proceed by finding comparable bounds for the coefficient and root entropy.

First we bound the coefficient entropy above. Each linear factor $(x - b_j)$ acts by convolution with the kernel $(1, b_j)$ whose support size is 2. This gives the inequality [CT06]:

$$S_c(p * (1, b_j)) - S_c(p) \leq \ln(1 + |b_j|) \leq \ln(1 + C).$$

Iterating over $j = 1, \dots, k$ gives

$$(3.2) \quad S_c(Y) - S_c(X) \leq \sum_{j=1}^k \ln(1 + C) = k \ln(1 + C).$$

Next, we bound the root entropy below. Notice that appending a single root $r = (x - b)$ acts as a convex combination. Let P be the normalized root multiplicity distribution of m elements, and let d be the single root distribution. Appending the new root is

$$Q = \left(1 - \frac{1}{m+1}\right)P \oplus \frac{1}{m+1}d.$$

This Shannon entropy H of the mixture is exactly [CT06]

$$H(Q) - H(P) = -\left(1 - \frac{1}{m+1}\right) \ln\left(1 - \frac{1}{m+1}\right) - \frac{1}{m+1} \ln\left(\frac{1}{m+1}\right) - \frac{1}{m+1} H(P).$$

Since $H(P)$ is bounded below by the uniform distribution on m elements, we find

$$\begin{aligned} H(Q) - H(P) &\geq -\left(1 - \frac{1}{m+1}\right) \ln\left(1 - \frac{1}{m+1}\right) - \frac{1}{m+1} \ln\left(\frac{1}{m+1}\right) - \frac{\ln m}{m+1} \\ &= \ln(m+1) - \frac{\ln m}{m+1} > \frac{\ln(m+1)}{m+1}. \end{aligned}$$

Now start at degree n with any multiplicity vector $P^{(n)}$. Append k new simple roots one at a time, obtaining vectors $P^{(n+1)}, \dots, P^{(n+k)}$. Summing the above bound for $m = n, n+1, \dots, n+k-1$ and using the fact that $\ln x/x$ decreases for $x > e$, yields

$$H(P^{(n+k)}) - H(P^{(n)}) \geq \sum_{j=0}^{k-1} \frac{\ln(n+j+1)}{n+j+1} \geq \frac{k}{n+k} \ln(n+k).$$

Returning to our original notation since $H(P) = S_r(p)$, we achieve a lower bound

$$(3.3) \quad S_r(pr) - S_r(r) \geq \frac{k}{n+k} \ln(n+k).$$

Comparing the bounds of Equation 3.2 and Equation 3.3, we find

$$S_c(pr) - S_c(p) \leq k \ln(1 + C) \leq \frac{k \ln(n+k)}{n+k} \leq S_r(pr) - S_r(p).$$

This gives that $\delta_c^r(f) \geq 0$ whenever

$$\ln(1 + C) \leq \frac{\ln(n+k)}{n+k}.$$

This inequality holds whenever $n+k \geq N$ where N solves the equation

$$\ln(1 + C) = \frac{\ln(N)}{N}.$$

□

Corollary 3.3. *The pair (S_c, S_r) exhibits a **closed phase transition** when morphisms meet the conditions of Proposition 3.2. The phase transition is visualized on a subset of objects and morphisms in Figure 2.*

Proof. It remains to construct a strict overlap to show that $A \neq A \cap B$ and $B \neq A \cap B$ through explicit polynomials. The numerical computations in Figure 2 reveal that $(x - 8)^{15}$ has strictly lower root entropy and $(x - 8)^5(x - 1)(x - 2)(x - 0.4234)(x - 0.5)(x - 0.1056)(x - 4)(x - 40)(x - 4)(x - 19)$ has strictly lower coefficient entropy. □

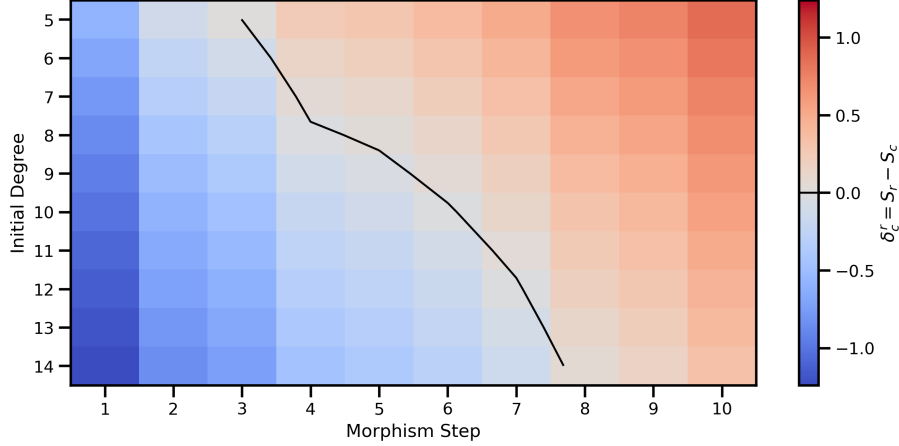


FIGURE 2. Phase transition in polynomial category for initial degrees $5 \leq n \leq 14$. Blue indicates root dominance, red coefficient dominance, and the black curve marks the phase boundary $\delta_c^r = 0$. Morphism sequences from Table 1 in Appendix A.

4. DISCUSSION

The two case studies illustrate complementary manifestations of closed phases. The quantum state example demonstrates how varying the width of a potential differently affects the informational content of position and momentum coordinates. At the critical width, $\lambda = 1.66$, information concentration is balanced. Narrower potentials focus position information while dispersing momentum, whereas wider potentials reverse this balance, highlighting a clear phase transition.

In the polynomial example, multiplication by polynomials with disjoint roots acts analogously to a noisy channel, increasing complexity and disorder with higher degrees and additional roots. Although the coefficient and root vectors fully determine the polynomial within the category, each basis gains a distinct informational advantage at different stages: the root representation initially concentrates information effectively, but after successive morphisms introduce additional disjoint roots, the coefficient representation becomes informationally dominant.

4.1. Future Work. Algorithms designed to detect and locate phase boundaries—for instance, to guide feature representation selection in machine learning—could significantly enhance computational scalability. The induced phase topology can direct efficient sampling strategies via covering families, allowing systematic exploration of the local-to-global structure inherent in phases. Modern geometric deep learning approaches and structurally constrained architectures inherently encode domain-specific geometric and topological assumptions, creating distinct decompositions of data through chosen feature spaces and parameterizations [BBCV21]. As data smoothly transitions between different underlying structures—such as from grid-like spatial data to irregular graph connectivity, or from linear relationships to polynomial dependencies—we expect the entropy of residuals between models with different inductive biases to undergo characteristic phase transitions. For instance, convolutional neural networks (designed for spatial locality) and graph neural networks (designed for arbitrary connectivity) should exhibit crossing points in their relative information concentration as data structure varies. Analyzing these phase transitions between geometric decompositions offers quantitative insights into how imposed architectural constraints simplify or complicate modeling tasks, providing principled criteria for model selection. Future computational methods could leverage this phase-topology structure for efficient selection of optimal feature representations based on data characteristics, significantly improving scalability and interpretability in data-intensive applications while avoiding the costly trial-and-error approach typically used in architecture selection.

Constructing sequences of decompositions, each with increasing informational advantage over the previous, could reveal multiscale structure analogous to renormalization group methods in physics. Such chains

naturally form topological towers, each refining the previous one. Variational principles from ergodic optimization or statistical mechanics might select optimal decompositions by extremizing entropy, creating bridges between foundational theory and practical applications in optimization, physics, and data analysis. Furthermore, the inherent categorical structure of phase topology would constrain permissible variations, ensuring consistency with the underlying mathematical framework.

One caveat is our inherent bias toward entropy minimization, reflecting a preference for tighter information compression. In contexts where entropy signifies uncertainty or diversity, higher entropy could instead be advantageous. In these scenarios, the dominance criteria can be inverted without altering the topological structure—only its interpretation changes. Clarifying these domain-specific interpretations will further integrate categorical entropy methods across mathematics, physics, and data science. The flexibility in comparing arbitrary decompositions, combined with the rich structure of phase topologies, positions this framework as a powerful tool for understanding complexity and information concentration across diverse disciplines.

APPENDIX A. POLYNOMIAL MORPHISMS

TABLE 1. Ten random morphisms chosen to make Figure 2.

Morphism Step	Appended Roots
1	1, 2
2	0.4234, 0.5, 0.1056
3	4
4	40, 4, 19
5	0.967
6	60
7	3, 5
8	7, 9, 11
9	1.456, 5.6
10	0.123, 0.456, 0.789, 0.1011

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